

Beyond the Prompt: Getting Started with Agentic AI for Geomatics Workflows



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Presentation Roadmap

- Getting Our Bearings: A Brief History of AI and Key Definitions
- Three Examples of Agentic Solutions:
 - Plot Vault – Using AI to Simplify CAD and Control Exports
 - Surveyor Sim – Using Agentic AI to Develop Educational Software
 - Survey Sleuth – Using Agentic AI to Search and Research Scanned Survey Records



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How Have You Used AI?

- I have never used AI
- I use AI to chat (*ChatGPT, Claude, Copilot, Gemini*)
- I use AI to co-work (*Claude Cowork, CoPilot Cowork*)
- I use Agentic AI (*OpenClaw, Claude Code, Hermes*)



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AI Didn't Start With ChatGPT



1950

Can Machines Think?



1956

AI Gets Its Name



1960s–1980s

Rules and Expert Systems



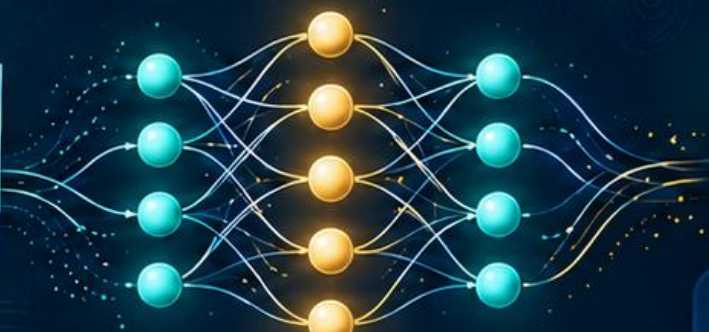
1990s–2000s

Prediction and Modeling



2012

Deep Learning Breakthrough



2017

The Transformer



2022

ChatGPT Opens
the Conversation



Today

From Conversation to Action

We are living through a revolution

- This technology is real and it is coming faster than you think.
- There is a global competition to implement AI everywhere.
- Any company stepping off the AI wheel doesn't slow the wheel, it just means that you are not on the wheel...



TBPELS Advisory Opinion on AI

- Policy Advisory Opinion 71, approved 14 November 2024.
 - Not a ban and not a new rule
- Three caveats
 - **Oversight** - Do not rely on AI output without review. You answer for anything you sign and seal.
 - **Competence** - Do not use AI on a project where you lack established competence.
 - **Confidentiality** - Know how the software uses and stores the client data you give it.
- *Responsible charge still governs*

AI Terms that you will hear today

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning
- Generative AI
- Large Language Model (LLM)
- Prompt
- Effort
- Token
- Context Window
- Hallucination
- AI Agent
- Agentic AI
- Skills
- Model Context Protocol (MCP)
- Version Control (Git)
- Markdown

Understanding AI - Key Terms

Artificial Intelligence (AI) – Computer systems associated with human intelligence

How AI Learns:

- *Machine Learning*: Learns patterns from data to predict
- *Deep Learning*: Uses neural networks to learn complex patterns

What AI Creates:

- *Generative AI*: Learns patterns to generate new content



How We Communicate with AI: LLMs and Prompts

Large Language Models (LLMs) – AI models trained on large amounts of text to process and generate language.

What LLMs Do:

- Answer questions, summarize documents, draft text, and generate code.

• How We Direct Them:

- **Prompts:** The questions, instructions, and context we provide to guide the model's response.
- **Effective Prompts:** Describe the task, supply relevant information, and specify the desired output.



What a large language model actually does

It predicts the next word. That's it.

- **Trained on text:** Billions of documents
- **Learns patterns:** Grammar, facts, logic, and code emerge
- **Scale surprised us:** Bigger models unlocked skills nobody explicitly programmed



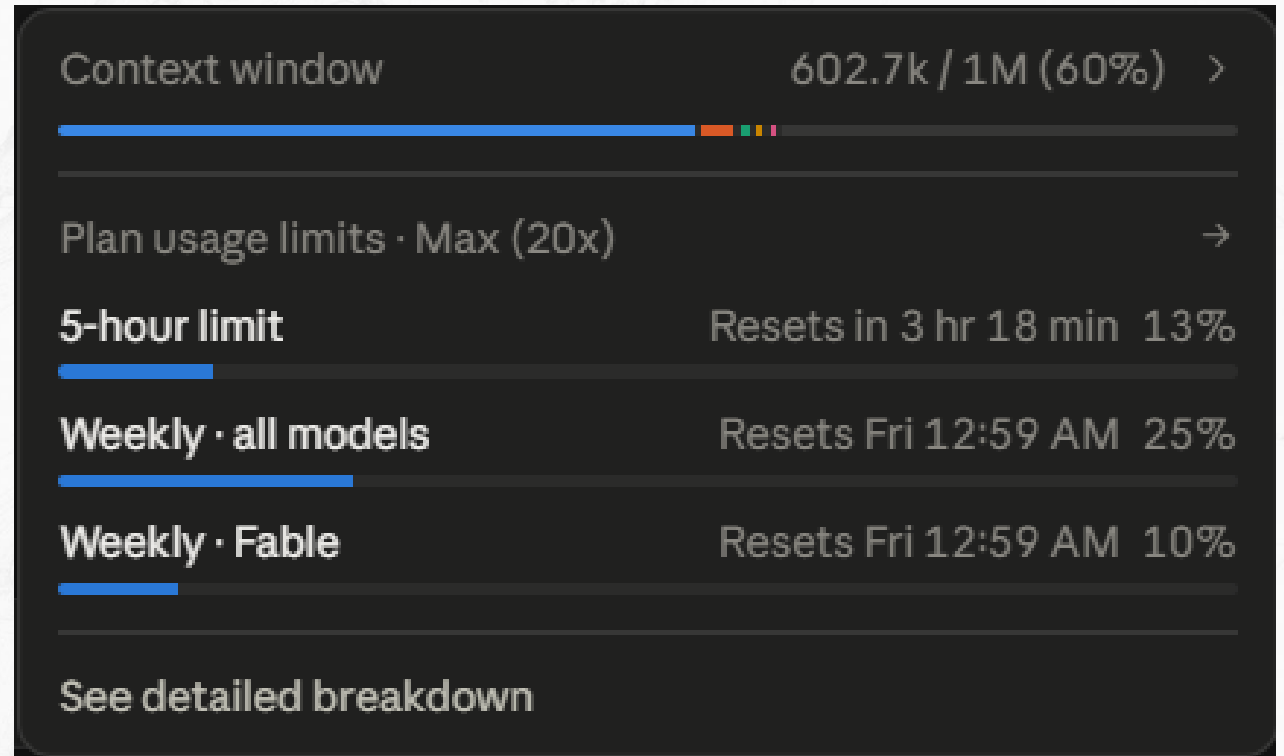
How AI Works

Effort: Reasoning

Context: Working Memory

Tokens: Chunks of text

Usage: Remaining usage



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How AI Can Go Wrong

Hallucination: When AI presents invented or incorrect information as factual.

It can happen when AI fills gaps with plausible content rather than acknowledging uncertainty.



Where it breaks and who owns the result



Confidently wrong

It can fabricate an API parameter or a 'fact' with a straight face. Plausible $\neq$ correct.



Security & data

Agents with tools can be tricked by malicious content. Least privilege; keep secrets out of prompts.



Cost & runaway loops

Token bills and infinite retries are real. Set budgets, timeouts, and step limits.



Accountability

The accountability is still yours. GISP, RPLS, PE, the agent did it is not a defense.

Treat agent output like a junior employee: trusted only after you check the work.

Chatbot → Copilot → Agent



Chatbot

Answers questions. You do all the work; the AI gives advice.



Copilot

Works alongside you. You drive each step; the AI drafts, assists, and can directly work on files on your computer.



Agent

Pursues a goal. You define the outcome; the AI plans, acts, and iterates. It creates solutions that you can direct and review.

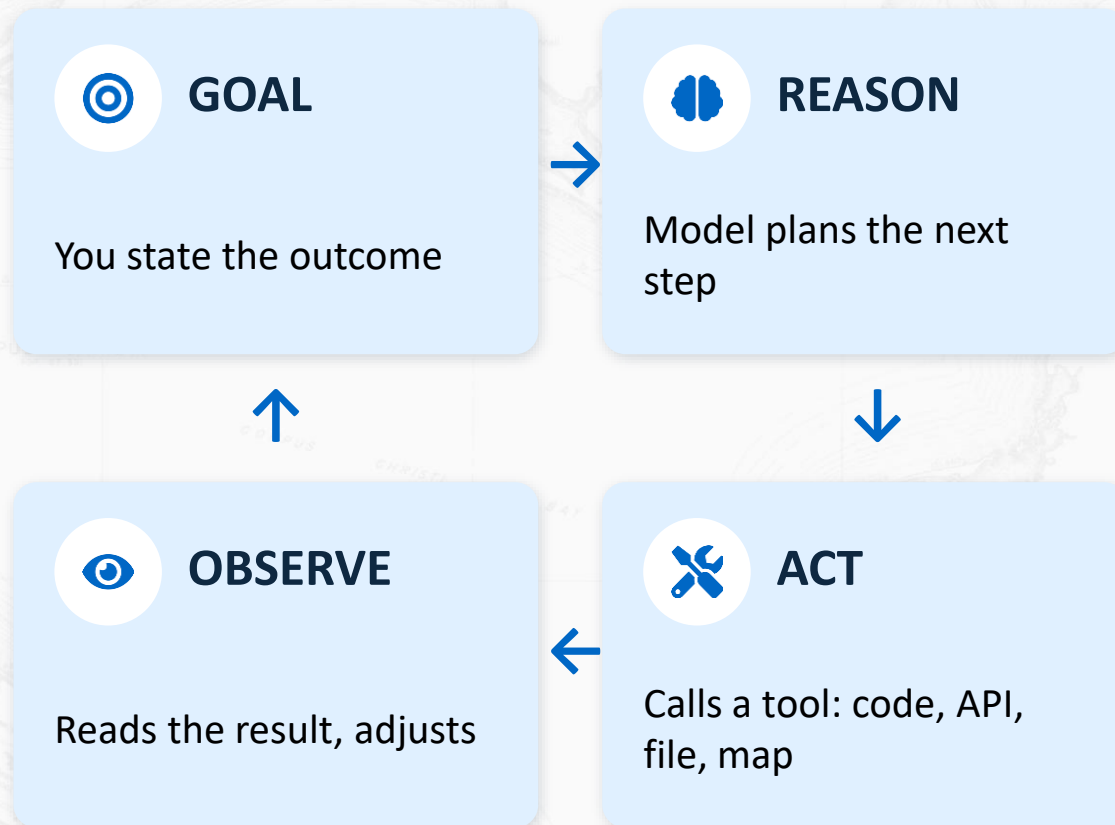
***The shift:** from generating text about work to actually doing the work.*



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The agentic loop



repeat until done — or until it asks for help

What the loop runs on

-  **A model**
The reasoning engine (Claude, GPT, Gemini...)
-  **Tools**
Terminal, Python, web, file system, GIS APIs
-  **Context & memory**
Your data, your instructions, what it's done so far
-  **Guardrails**
Permissions: what it may touch, and what it must ask about

How Agentic AI Takes Action

AI Agent: A system that uses AI and tools to complete tasks.

Agentic AI: AI that plans, acts, and adjusts its approach toward a goal.

Skills: Reusable instructions and resources for specific tasks.

Model Context Protocol (MCP): A standard for connecting AI applications to tools and data.



How We Organize and Track the Work

- **Version Control (Git):** Tracks changes to project files so you can compare versions, collaborate, and restore earlier work.
- **Markdown:** A plain-text format that uses simple symbols to organize headings, lists, links, and other content.
- **Why They Matter:**
 - Git records what changed. This is your version control and undo button.
 - Markdown keeps instructions and documentation easy for people and AI to read. This is your documentation.



Markdown

- Lightweight markup language
- Human-readable syntax
- Converts to HTML
- Universal developer standard – AI *loves* markdown
- Portable and future-proof



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Collector redesign – Phase 3 (the shell), broken into steps

****Rick, on Phase 1: "Can we split this into more tickets/steps so we take it carefully?"**** It worked – ten tickets, three waves, all shipped. Same split here, and Phase 3 needs it more: this is the ****first phase a student will see****. Phases 0, 1 and 2 were model and maths. If they were wrong, a test went red. If this is wrong, someone cannot take a shot in a lab.

First live class: ****2026-09-08****.

⚠ THE PLAN'S RELEASE VALVE FOR THIS PHASE DOES NOT EXIST

`docs/collector-redesign-plan.md` §9 mitigates **"One app is a rewrite, not a merge"** with ****"Knob off (Phase 0) so it can ship unfinished without reaching a class."**** That is false, verified three ways:

```
| | |
|---|---|
| `collectorPanel.setAvailable()` call sites | **one** – `src/main.ts:8649` |
| callers of the `toolsConfigure` hook it sits in | **one** – `src/classroom/assignmentSession.ts:1461` |
| free-play boots reaching either | **zero** |
| `#hud-collector` in the markup | no `hidden`, no `display:none` – **visible by default** |
| production free-play HTML | the button is there |
```

The knob is an ****assignment-mode hook****. In the sandbox the collector is always available, and surveyorsim.com is public. ****A shell shipped under this plan's stated mitigation reaches every visitor the day it deploys**** – Rick included, when he tests.

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Three Problems Solved with Agentic AI



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Problem Statement 1

“I have a huge CAD file that holds all surveys we have done in a county and it takes 30 seconds to regenerate every time we pan or zoom. We need to extract CAD data and control several times a day from this file and it takes us way too long to do this because of the slow speed.”



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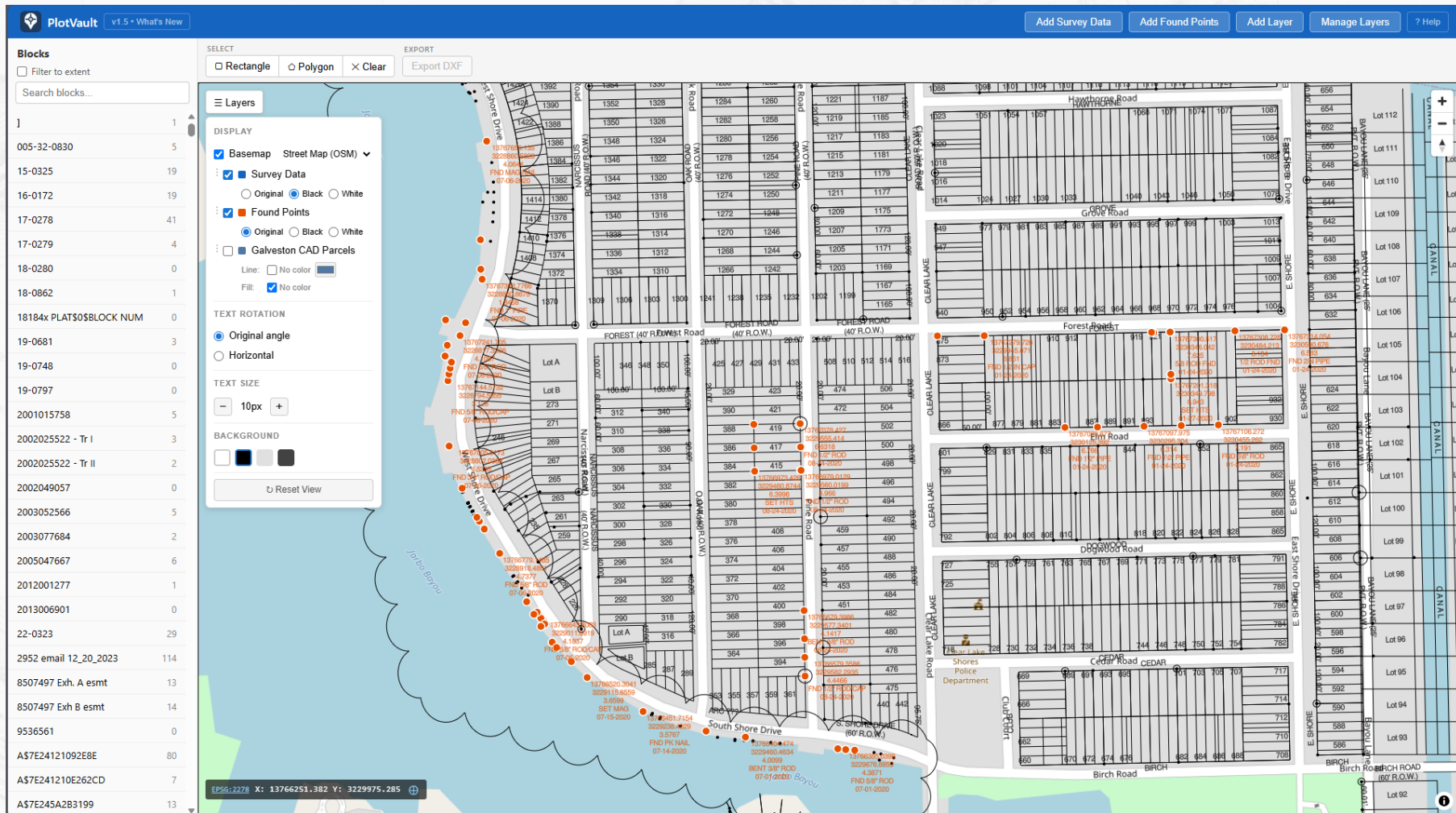
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Problem Statement 1

- What problem were we solving?
 - Taking time to render in AutoCAD
 - Only one person could have the file open at a time
- How did Agentic AI Help?
 - Wrote a small bespoke software that did exactly what was needed and runs locally within the office



Plot Vault



Problem Statement 2

“We teach online surveying labs and want a simulation of surveying so that online students can learn and experiment. Software installation is often a challenge at schools, so we want something that will run in a web browser with no installation.”



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Problem Statement 2

- What problem were we solving?
 - Online surveying simulator for teaching and learning
 - Software installation can be a barrier to use
- How did Agentic AI Help?
 - Wrote web-based software to provide a simulator for both students and instructors. Models physics, weather, and measurements as well as field collection equipment.



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SurveyorSim.com



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Problem Statement 3

“I have thousands PDFs of old scanned surveys and documents that I have not had time to sort through and geolocate. Additionally, even if I do locate these, I want to link them to plats and the county CAD, if able.”



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Survey Sleuth

SurveySleuth
1518 Beachtown Dr, Galveston, TX 77550

within 1/2 mi
Search

Parcel ID

2

10-1627

31 ft

1541 Beachtown Drive, Galveston

Elevation Certificate

1 Record

placed by Parcel ID

Zone VE - BFE 15 ft - floor 7.3 ft - ground 5.7 ft - Construction drawings - Mar 2011

1

10-0024

512 ft

719 Shiraz Passage, Galveston

Elevation Certificate

1 Record

certificate lat/long, checked by Parcel ID

Zone VE - BFE 16 ft - floor 6.6 ft - ground 6.4 ft - Building under construction - Jan 2010

4

11-0209

512 ft

1625 East Beach Dr., Galveston

Elevation Certificate

1 Record

certificate lat/long, checked by Parcel ID

Zone VE - BFE 16 ft - floor 6.4 ft - ground 6.0 ft - Building under construction - Feb 2011

6

11-0971

512 ft

1625 East Beach Dr., Galveston

Elevation Certificate

Survey

3 Records

11-0971-REV.pdf: certificate lat/long, checked by Parcel ID

11-0971.pdf: certificate lat/long, checked by Parcel ID

11-0971.pdf: placed by Parcel ID, since split

Parcel split since this Record. Placed at the centre of its pieces.

Elevation Certificate

11-0971-REV.pdf

Jul 14, 2011 - 4 pages

Zone VE - BFE 16 ft - floor 6.4 ft - ground 6.0 ft - Finished construction - Jul 2011

Elevation Certificate

11-0971.pdf

no date - 4 pages

Zone VE - BFE 16 ft - floor 6.4 ft - ground 6.0 ft - Building under construction

Survey

11-0971.pdf

Jul 7, 2011 - 1 page

Lot 6, Block 8, REPLAT BEACHTOWN GALVESTON, VILLAGE ONE, SECTION ONE - BLOCKS 7 & 8 AND BEACHTOWN GALVESTON, VILLAGE ONE, SECTION TWO - BLOCK 1, LOTS 1-4, BLOCKS 2-5, AND BLOCK 6, LOTS 1-4 - In the 100-year flood plain - Jul 2011

6

11-1814

831 ft

1401 East Beach Drive, Galveston

Elevation Certificate

Survey

2 Records

11-1814.pdf: placed by Parcel ID

11-1814.pdf: certificate lat/long, checked by Parcel ID

Grand Beach - In the 100-year flood plain - Dec 2011

7

10-1463

848 ft

1549 Seaside Dr., Galveston

Elevation Certificate

1 Record

certificate lat/long, checked by Parcel ID

Zone VE - BFE 19 ft - floor 7.5 ft - ground 6.5 ft - Construction drawings - Oct 2010

1

10-0053

939 ft

1813 E. Beach Dr., Galveston

Elevation Certificate

2 Records

10-0053-REV-ALL-5.pdf: certificate lat/long, checked by address

10-0053.pdf: certificate lat/long, checked by Parcel ID

Zone VE - BFE 16 ft - floor 6.3 ft - ground 5.6 ft - Finished construction - Jan 2010

1

10-0054

939 ft

710 Positano Rd., Galveston

Elevation Certificate

1 Record

certificate lat/long, checked by Parcel ID

Zone VE - BFE 16 ft - floor 6.3 ft - ground 5.6 ft - Finished construction - Jan 2010

Map

Record: 11-0971.pdf

+

-

● Certificate

● Survey

● Both

1000 ft

Leaflet

© OpenStreetMap contributors

SOURCES

1

Elevation Certificate

10-1420.pdf

Job 10-1420 - Oct 1, 2010

2

Elevation Certificate

10-1627.pdf

Job 10-1627 - Mar 8, 2011

3

Elevation Certificate

10-0024.pdf

Job 10-0024 - Jan 6, 2010

4

FEMA

FEMA National Flood Hazard Layer

5

Elevation Certificate

11-0209.pdf

Job 11-0209 - Feb 24, 2011

6

NGS

NGS Benchmarks

7

GCAD

Galveston CAD parcels

1642-0000-0000-001, 1642-0000-0000-002, 1642-0000-0000-005, 1642-0000-0000-010, 1642-0000-0000-014, 1642-0000-0000-015, 1642-0000-0000-016, 1642-0000-0000-017, 1642-0000-0000-018, 1642-0000-0000-019, 1642-0000-0000-020, 1642-0000-0000-021

8

GCAD

Galveston CAD parcels

1646-0000-0000-002, 1646-0000-0000-004, 1646-0000-0000-005, 1646-0000-0000-007, 1646-0000-0000-008

9

GCAD

Galveston CAD parcels

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Build Survey Sleuth at your Company



<https://ricksmith.github.io/SurveySleuth-Starter/>



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Development Cycle

- Plan
- Specification
- Split into Workable Chunks
- Implement
- Review/Test
 - Unit, End to End, and Manual
- Deploy



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Takeaways

- Don't hurry through this process.
- Let your agent earn responsibility one step at a time.
- Agents are multipliers.
 - Make expectations clear so you don't multiply by zero.
- Conduct experiments. Have fun!
- If you don't understand something, ask the AI!



Questions?

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ricksmith.github.io/SurveySleuth-Starter